

Rayan Hossain

Computer Science and Engineering Student — AI & Machine Learning Research Enthusiast

📍 Rajshahi, Bangladesh ✉️ rayanhossain239@gmail.com ☎️ 01532523297
🌐 linkedin.com/in/ryan-hossain-41337b24b 🐙 github.com/ryan239

SUMMARY

Computer Science and Engineering undergraduate at Rajshahi University of Engineering & Technology (RUET) with research experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing. Currently serving at ELITE Research Lab, contributing to AI-driven research and experimental development. First Author of an IEEE conference paper on attention-enhanced image classification. Skilled in developing and evaluating deep learning models, conducting research experiments, and transforming research ideas into practical solutions. Passionate about Large Language Models (LLMs), Multimodal AI, Computer Vision, and intelligent systems.

EDUCATION

BSc in Computer Science and Engineering

Rajshahi University of Engineering & Technology (RUET)

CGPA: **3.88 out of 4.00**

7th Semester

Dec 2022 – Current

Higher Secondary Certificate (HSC)

National Ideal College

GPA: **5.00 out of 5.00**

2021

Secondary School Certificate (SSC)

National Ideal School, Dhaka

GPA: **5.00 out of 5.00**

2019

RESEARCH EXPERIENCE

ELITE Research Lab

2026 – Present

- Contributing to ongoing research projects in Artificial Intelligence and Machine Learning.
- Assisting in model development, experimentation, performance evaluation, and research documentation.
- Collaborating with researchers on computer vision and natural language processing applications.
- Developing practical research skills in scientific investigation, data analysis, and AI-driven problem solving.

TECHNICAL SKILLS

Programming Languages

Python, C, C++, Java, JavaScript, Assembly

Machine Learning & AI

Supervised Learning, Deep Learning, Transfer Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Neural Networks

ANN, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms

Deep Learning Concepts

Activation Functions, Loss Functions, Optimizers, Back-propagation, Batch Normalization, Regularization, Overfitting Mitigation, Model Fine-Tuning

Computer Vision

Image Classification, Object Detection Fundamentals, Image Processing, Data Augmentation, Transfer Learning

Natural Language Processing

Text Classification, Emotion Detection, Multilingual NLP, Embeddings, Transformer Architectures, Large Language Models

Frameworks & Libraries

TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

Tools & Platforms

Git, GitHub, Linux (Ubuntu), MySQL, VS Code, Overleaf, LaTeX

RESEARCH INTERESTS

- Artificial Intelligence
- Machine Learning and Deep Learning
- Large Language Models (LLMs)
- Natural Language Processing
- Computer Vision
- Multimodal AI
- Intelligent Systems

PUBLICATION

EfficientNetB3 with CBAM and Multi-Scale Feature Fusion for High-Accuracy Garbage Image Classification

Conference: International Conference on Electrical, Computer and Telecommunication Engineering (ICECTE) 2026

Venue: Rajshahi University of Engineering & Technology (RUET)

Paper ID: 1041

Status: Accepted for presentation and publication

IEEE Link:

ieeexplore.ieee.org/document/11429240

DOI:

10.1109/ICECTE69292.2026.11429240

Drive Link:

Google Drive Paper Link

Contribution: Complete research paper written independently, including problem formulation, literature review, methodology design, implementation, experimentation, result analysis, and final manuscript preparation.

ONGOING RESEARCH

Contrastive Learning for Language-Invariant Emotion Detection

- Contributing to multilingual emotion detection research involving Bangla, Banglish, and English text.
- Exploring contrastive learning, knowledge distillation, and transformer-based architectures.
- Investigating efficient language-invariant representations for low-resource multilingual NLP applications.

PROJECTS

SmartWasteVision – AI-Powered Garbage Classification and Smart Recycling Management System

- Developed an AI-driven web application for automated garbage classification and smart recycling management.
- Worked with image preprocessing, normalization, resizing, augmentation, model training, evaluation, and optimization.
- Applied deep learning concepts for image-based waste classification.

A Comprehensive Car Marketplace Platform

- Built a modern and responsive car marketplace platform using React, Clerk Authentication, and Tailwind CSS.

- Implemented user-friendly browsing, searching, and car listing features.

Personal Portfolio Website

- Designed a simple personal portfolio using HTML and CSS.

Flappy-Bird Game

- Developed a 2D game using HTML, Dart, C++, CMake, and Swift.

Plane Shooting Game

- Created a basic plane shooting game using Python.

RESEARCH & PRACTICAL EXPERIENCE

- Worked with large image datasets for deep learning-based classification tasks.
- Applied preprocessing techniques such as image resizing, normalization, and data augmentation.
- Gained practical experience in model training, performance evaluation, and optimization.
- Used EfficientNetB3, CBAM, and Multi-Scale Feature Fusion for high-accuracy image classification.
- Prepared academic research manuscript, including literature review, methodology, experimental results, and discussion.

ACHIEVEMENTS

- First Author of an IEEE Conference Paper (ICECTE 2026)
- Research Intern at Code Studio AI Research Lab
- CGPA: 3.88/4.00 at RUET
- Codeforces Maximum Rating: 1039
- Solved 190+ Algorithmic Problems on Codeforces

LANGUAGES

- Bangla
- English

ADDITIONAL INFORMATION

- Open to AI/ML research, internship, and software development opportunities.
- Reference available upon request.